Quantum Scattering Theory: Partial Wave Analysis

Recall from the hydrogen atom that the wavefunction can be separated

$$\psi(r,\theta,\phi) = R(r)Y_\ell^m(\theta,\phi)$$

With $Y_\ell^m(\theta,\phi)$ the spherical harmonics and $R(r)$ the radial equation

We can use $u(r) = rR(r)$ to get the radial Schrodinger equation

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2}\right]u = Eu$$

At large r the potential $V(r)$ and the centrifugal force $\frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2}$ will both be negligible and so we have

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} = Eu \longrightarrow \frac{d^2u}{dr^2} = -\frac{2mE}{\hbar^2}u$$

With $k = \frac{\sqrt{2mE}}{\hbar}$ we then have that $k^2 = \frac{2mE}{\hbar^2}$ and so

$$\frac{d^2u}{dr^2} = -k^2u$$

The general solution for $u(r)$ is

$$u(r) = Ce^{ikr} + De^{-ikr}$$

Since De^{-ikr} is an incoming wave and we are interested in the outgoing (scattered) wave, we set $D = 0$ and, since $u(r) = rR(r) \longrightarrow R(r) = \frac{u(r)}{r}$ we get for very large r that $R(r) \sim \frac{e^{ikr}}{r}$

We have three regions we need to work with

The region inside the potential is

$$V(r) \neq 0 \qquad \frac{\ell(\ell+1)}{r^2} \neq 0$$

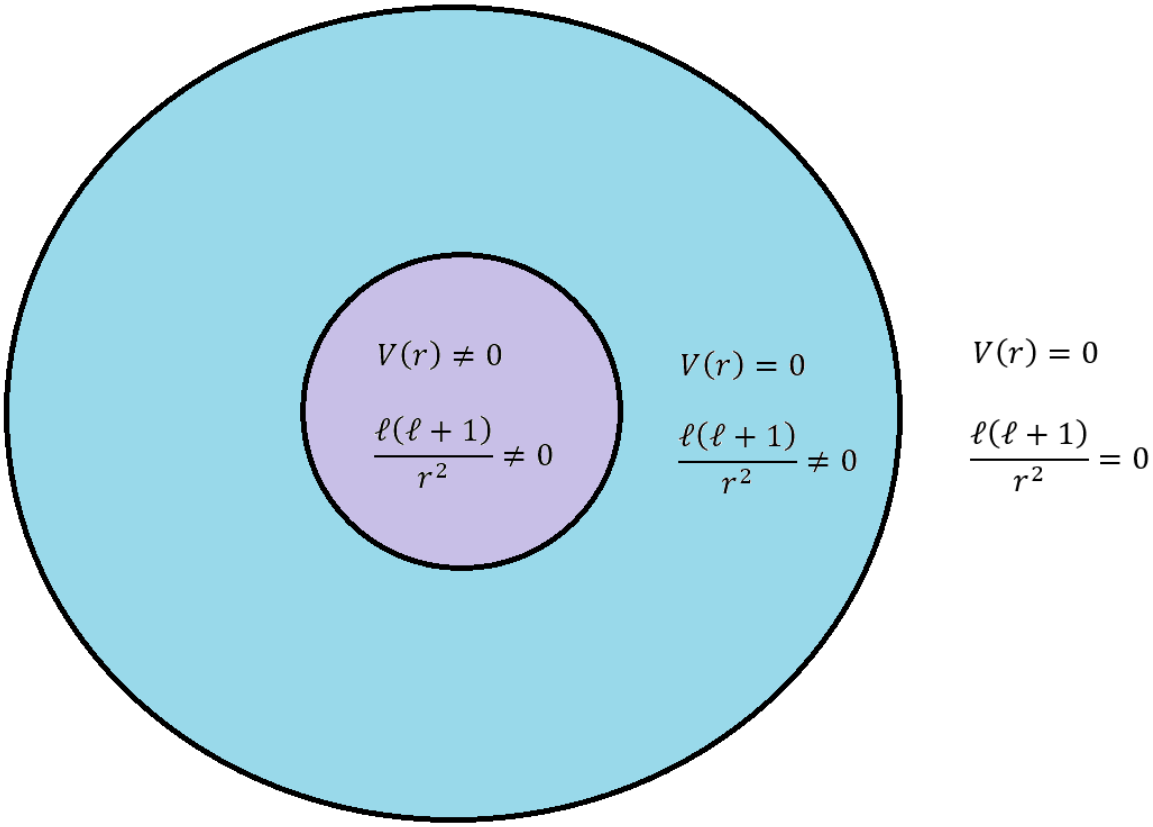
The intermediate region is

$$V(r) = 0 \qquad \frac{\ell(\ell+1)}{r^2} \neq 0$$

And the distant region, when r is large, is

$$V(r) = 0 \qquad \frac{\ell(\ell+1)}{r^2} = 0$$

Where we will need the wavefunction to be a superposition of the incoming plane wave and the outgoing spherical wave, i.e., $\psi(r,\theta) = \left(e^{ikz} + f(\theta)\frac{e^{ikr}}{r}\right)$



In the intermediate region, where $V(r) \approx 0$ and $\frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2} \neq 0$ we have

$$-\frac{\hbar^2}{2m}\frac{d^2r}{dr^2} + \frac{\hbar^2}{2m}\frac{\ell(\ell+1)}{r^2} = Eu$$

$$\frac{d^2r}{dr^2} - \frac{\ell(\ell+1)}{r^2} = -k^2u$$

For this the general solution is the **spherical Hankel functions**

$$h_\ell^{(1)}(x) = j_\ell(x) + in_\ell(x) \qquad h_\ell^{(2)}(x) = j_\ell(x) - in_\ell(x)$$

Where $j_\ell(x)$ and $n_\ell(x)$ are the spherical Bessel functions

$j_0(x) = \frac{\sin(x)}{x}$	$n_0(x) = -\frac{\cos(x)}{x}$
$j_1(x) = \frac{\sin(x)}{x^2} - \frac{\cos(x)}{x}$	$n_1(x) = -\frac{\cos(x)}{x^2} - \frac{\sin(x)}{x}$
$\vdots$	$\vdots$
$j_\ell(x) = (-x)^\ell \left(\frac{1}{x}\frac{d}{dx}\right)^\ell \frac{\sin(x)}{x}$	$n_\ell(x) = -(-x)^\ell \left(\frac{1}{x}\frac{d}{dx}\right)^\ell \frac{\cos(x)}{x}$

Using the trigonometric identities

$$e^{ix} = \cos(x) + i\sin(x) \qquad e^{-ix} = \cos(x) - i\sin(x)$$

The spherical Hankel functions of the first and second kind analogous to e^{ikr} and e^{-ikr} respectively.

$h_0^{(1)}(x) = -i\frac{e^{ix}}{x}$	$h_0^{(2)}(x) = i\frac{e^{-ix}}{x}$
$h_1^{(1)}(x) = \left(-\frac{i}{x^2} - \frac{1}{x}\right)e^{ix}$	$h_1^{(2)}(x) = \left(\frac{i}{x^2} - \frac{1}{x}\right)e^{-ix}$
$h_2^{(1)}(x) = \left(-\frac{3i}{x^3} - \frac{3}{x^2} + \frac{i}{x}\right)e^{ix}$	$h_2^{(2)}(x) = \left(\frac{3i}{x^3} - \frac{3}{x^2} - \frac{i}{x}\right)e^{-ix}$
$\vdots$	$\vdots$
$h_\ell^{(1)}(x) = \frac{1}{x}(-i)^{\ell+1}e^{ix}$	$h_\ell^{(2)}(x) = \frac{1}{x}(i)^{\ell+1}e^{-ix}$

And so, for the outgoing wave we will use for the radial equation the Hankel function of the first kind

$$R(r) = h_\ell^{(1)}(kr)$$

Since we are assuming the potential is spherically symmetric, meaning that it does not depend on ϕ, then our quantum number $m = 0$ and so

$$Y_\ell^0(\theta, \phi) = \sqrt{\frac{2\ell + 1}{4\pi}} P_\ell(\cos\theta)$$

Where $P_\ell(\cos\theta)$ is the ℓth Legendre polynomial. We therefore have

$$\psi(r, \theta, \phi) = A\left[e^{ikz} + \sum_{\ell,m} C_{\ell,m} h_\ell^{(1)}(kr) Y_\ell^m(\theta, \phi)\right]$$

First term in square brackets is incoming plane wave; second is superposition of outgoing spherical waves

With our $Y_\ell^0 = \sqrt{\frac{2\ell+1}{4\pi}} P_\ell(\cos\theta)$ we can then redefine the expansion coefficients with $m = 0$ to become $C_{\ell,0} \equiv i^{\ell+1}k\sqrt{4\pi(2\ell + 1)}a_\ell$ (where a_ℓ is called the **partial wave amplitude**) and then get

$$C_{\ell,0}Y_\ell^0 = i^{\ell+1}k\sqrt{4\pi(2\ell + 1)}a_\ell \sqrt{\frac{2\ell + 1}{4\pi}} P_\ell(\cos\theta) = (2\ell + 1)i^{\ell+1}ka_\ell P_\ell(\cos\theta)$$

Which then gives us

$$\psi(r, \theta) = A\left[e^{ikz} + k\sum_{\ell=0}^{\infty} i^{\ell+1}(2\ell + 1)a_\ell h_\ell^{(1)}(kr) P_\ell(\cos\theta)\right]$$

At very large r (where $V(r) = 0$ and $\frac{\ell(\ell+1)}{r^2} = 0$, often called the **radiation zone**) our Hankel function of the first kind will converge on $h_\ell^{(1)}(kr) \xrightarrow{r\gg1} (-i)^{\ell+1}\frac{e^{ikr}}{kr}$ since the higher order terms of r will go to zero, and so will look like

$$h_\ell^{(1)}(kr) = (-i)^{\ell+1}\frac{e^{ikr}}{kr}$$

Which is a spherically symmetric wave and so we can put this into our solution for the intermediate region (thereby connecting the intermediate region with the radiation zone)

$$\psi(r,\theta) = A\left[e^{ikz} + \cancel{k}\sum_{\ell=0}^{\infty} i^{\ell+1}(2\ell+1)a_\ell(-i)^{\ell+1}\frac{e^{ikr}(kr)}{\cancel{kr}}P_\ell(\cos\theta)\right]$$

$$= A\left[e^{ikz} + \sum_{\ell=0}^{\infty}\underbrace{i^{\ell+1}(-i)^{\ell+1}}_{=1}(2\ell+1)a_\ell\frac{e^{ikr}(kr)}{r}P_\ell(\cos\theta)\right]$$

The $\frac{e^{ikr}(kr)}{r}$ is not dependent on ℓ and so can come out of the summation

$$= A\left[e^{ikz} + \frac{e^{ikr}(kr)}{r}\underbrace{\sum_{\ell=0}^{\infty}(2\ell+1)a_\ell P_\ell(\cos\theta)}_{=f(\theta)}\right]$$

And so, our $\psi(r,\theta) = A\left[e^{ikz} + \frac{e^{ikr}(r)}{r}f(\theta)\right]$ has for our scattering factor $f(\theta)$

$$\boxed{f(\theta) = \sum_{\ell=0}^{\infty}(2\ell+1)a_\ell P_\ell(\cos\theta)}$$

Which means that the **differential cross section is**

$$\boxed{D(\theta) = |f(\theta)|^2 = \sum_{\ell'}^{\infty}\sum_{\ell=0}^{\infty}(2\ell+1)(2\ell'+1)a_\ell a_\ell^* P_\ell(\cos\theta)P_{\ell'}(\cos\theta)}$$

And using that the Legendre polynomials $P_\ell(\cos\theta)$ are orthogonal, we need that $P_{\ell'}(\cos\theta) = P_\ell(\cos\theta)$ and

$$\int_{-1}^{1} P_\ell(\cos\theta)P_{\ell'}(\cos\theta)\,d\cos\theta = \frac{1}{2\ell+1}\delta_{\ell,\ell'}$$

And so, if $\ell \neq \ell'$ then it goes to zero. Likewise, we get

$$(2\ell+1)(2\ell'+1) = (2\ell+1)^2$$

And so

$$(2\ell+1)^2\left(\frac{1}{2\ell+1}\right) = 2\ell+1$$

Leaving us with

$$D(\theta) = |f(\theta)|^2 = \sum_{\ell=0}^{\infty} (2\ell + 1)|a_\ell|^2$$

And then with

$$D(\theta) = \frac{d\sigma}{d\Omega}$$

$$\frac{d\sigma}{d\Omega} = \sum_{\ell=0}^{\infty} (2\ell + 1)|a_\ell|^2$$

$$d\sigma = \sum_{\ell=0}^{\infty} (2\ell + 1)|a_\ell|^2 \, d\Omega$$

$$\int d\sigma = \int \sum_{\ell=0}^{\infty} (2\ell + 1)|a_\ell|^2 \, d\Omega$$

Recall that $d\Omega = \sin(\theta)\, d\theta\, d\phi$ and $D(\theta) = \frac{b}{\sin(\theta)} \left|\frac{db}{d\theta}\right|$ so

$$D(\theta) d\Omega = b\, db\, d\phi \qquad b \propto V(r)$$

Which means we must integrate over ϕ, which will be 4π which gives us the **total cross-section**

$$\sigma = 4\pi \sum_{\ell=0}^{\infty} (2\ell + 1)|a_\ell|^2$$

Now we then need to solve Schrodinger equation for the internal region (where $V(r) \neq 0$ and $\frac{\ell(\ell+1)}{r^2} \neq 0$) and match it to the exterior solution (blue boxed equation above)

But first we need to get the incident wave into spherical coordinates. To do this we use Rayleigh's formula

$$e^{ikz} = \sum_{\ell=0}^{\infty} i^\ell (2\ell + 1) j_\ell(kr) P_\ell(\cos\theta)$$

So, we can substitute this into the intermediate region (blue-boxed) equation to get

$$\psi(r,\theta) = A\left[\sum_{\ell=0}^{\infty} i^\ell (2\ell + 1) j_\ell(kr) P_\ell(\cos\theta) + k \sum_{\ell=0}^{\infty} i^{\ell+1} (2\ell + 1) a_\ell h_\ell^{(1)}(kr) P_\ell(\cos\theta)\right]$$

$$\psi(r,\theta) = A\left[\sum_{\ell=0}^{\infty} k i^{\ell+1} (2\ell + 1) a_\ell h_\ell^{(1)}(kr) P_\ell(\cos\theta) + i^\ell (2\ell + 1) j_\ell(kr) P_\ell(\cos\theta)\right]$$

$$\psi(r,\theta) = A \sum_{\ell=0}^{\infty} (2\ell + 1) i^\ell \left[ika_\ell h_\ell^{(1)}(kr) + j_\ell(kr)\right] P_\ell(\cos\theta)$$

Example

If we assume a hard-sphere scattering and a sphere of radius $r = a$ we have

$$V(r) = \begin{cases} \infty, & (r \le a) \\ 0, & (r \ge a) \end{cases} \qquad \psi(a, 0) = 0$$

Meaning that

$$\psi(r, \theta) = A \sum_{\ell=0}^{\infty} (2\ell + 1) i^{\ell} \left[ika_{\ell} h_{\ell}^{(1)}(ka) + j_{\ell}(ka) \right] P_{\ell}(\cos\theta) = 0$$

For all θ. But $(2\ell + 1) i^{\ell} \neq 0$ and $P_{\ell}(\cos\theta) \neq 0$, which means that

$$ika_{\ell} h_{\ell}^{(1)}(ka) + j_{\ell}(ka) = 0$$

$$ika_{\ell} h_{\ell}^{(1)}(ka) = -j_{\ell}(ka)$$

$$a_{\ell} = -\frac{j_{\ell}(ka)}{ikh_{\ell}^{(1)}(ka)}$$

Using that $\frac{1}{i} = -i$ we get

$$a_{\ell} = i\frac{j_{\ell}(ka)}{kh_{\ell}^{(1)}(ka)}$$

Where the i can be pulled out in the equation

$$A \sum_{\ell=0}^{\infty} (2\ell + 1) i^{\ell} \left[i\frac{j_{\ell}(ka)}{kh_{\ell}^{(1)}(ka)} \right] P_{\ell}(\cos\theta) = A \sum_{\ell=0}^{\infty} (2\ell + 1) i^{\ell+1} \left[\frac{j_{\ell}(ka)}{kh_{\ell}^{(1)}(ka)} \right] P_{\ell}(\cos\theta)$$

And so

$$\sigma = 4\pi \sum_{\ell=0}^{\infty} (2\ell + 1) |a_{\ell}|^2 = 4\pi \sum_{\ell=0}^{\infty} (2\ell + 1) \left| \frac{j_{\ell}(ka)}{kh_{\ell}^{(1)}(ka)} \right|^2$$

$$\sigma = \frac{4\pi}{k^2} \sum_{\ell=0}^{\infty} (2\ell + 1) \left| \frac{j_{\ell}(ka)}{h_{\ell}^{(1)}(ka)} \right|^2$$

If we look at when $ka \ll 1$ (the wavelength is much larger than the sphere, i.e., a low energy particle) then $n_{\ell}(z) \gg j_{\ell}(z)$ for small z and so

$$\frac{j_{\ell}(z)}{h_{\ell}^{(1)}(z)} \approx \frac{j_{\ell}(z)}{j_{\ell}(z) + in_{\ell}(z)} \approx -i\frac{j_{\ell}(z)}{n_{\ell}(z)}$$

$$\approx -i\frac{\left[\frac{2^{\ell}\ell!\, z^{\ell}}{2(\ell + 1)!}\right]}{\left[\frac{-(2\ell)!\, z^{-\ell-1}}{2^{\ell}\ell!}\right]} \approx \frac{i}{2\ell + 1}\left[\frac{2^{\ell}\ell!}{(2\ell)!}\right]^2 z^{2\ell+1}$$

Giving

$$\sigma \approx \frac{4\pi}{k^2}\sum_{\ell=0}^{\infty}\frac{i}{2\ell+1}\left[\frac{2^{\ell}\ell!}{(2\ell)!}\right]^4 (ka)^{4\ell+2}$$

With $ka \ll 1$ the higher powers of $(ka)^{4\ell+1}$ are negligible and so the scattering is mostly the $\ell = 0$ term

$$\frac{4\pi}{k^2}\frac{i}{1}\left[\frac{1}{1}\right]^4 (ka)^2 = i4\pi a^2$$

Which is on the order of

$$\sigma \approx 4\pi a^2$$

Which is the formula for the surface area of a sphere

This means that the particle is not just scattering off the face of the sphere pointing in the direction of the incident particle, but is scattering off *the entire sphere*!